School of Computer Sciences & Engineering

Department of Computer Science & Application

Capstone Project Synopsis On

'AI-Driven Predictive Analytics on Student Academic Performance and Tool Usage Patterns'

By:

Ms. Mahima Chaudhari

Class & Semester:

SY MCA Semester III

Under the Guidance of:

Prof. Smita Dhawadekar

Academic Year: 2026-27 (Odd Semester)

TITLE OF THE PROJECT: AI-Driven Predictive Analytics on Student Academic Performance and Tool Usage Patterns Using Machine Learning and Power BI

ABSTRACT:

The integration of Artificial Intelligence (AI) tools in higher education has grown exponentially, reshaping how students complete assignments, solve coding challenges, and prepare for academic evaluations. While tools such as ChatGPT, Google Gemini, and GitHub Copilot offer substantial assistance, their actual impact on academic performance specifically student grade point averages and conceptual retention remains a topic of critical discussion. Traditional academic surveys often provide static, descriptive snapshots of student feedback without modeling the underlying non-linear interactions between usage duration, trust levels, internet infrastructure, and academic outcomes.

To address this gap, the proposed system presents an end to end data intelligence pipeline that combines Machine Learning with Business Intelligence (BI) visualization. Using a dataset of 3,614 student records, the system processes categorical indicators (such as academic stream, primary AI tool, institutional policies, and device access) along with numerical variables (daily usage hours, trust rating, and awareness level). Feature preprocessing is executed using Scikit-Learn's ColumnTransformer, OneHotEncoder, and StandardScaler. A 4-layer Sequential Neural Network constructed via TensorFlow/Keras trains on these preprocessed vectors using the Adam optimizer and Mean Squared Error (MSE) loss function to predict student grade impact (Predicted_Impact_on_Grades). Finally, the output predictions and model error metrics (Absolute_Error) are exported into Microsoft Power BI Desktop to build an interactive, multi-dimensional executive dashboard. This project bridges data science, deep learning, and interactive data visual analytics to provide educational institutions with objective, data-driven insights.

INTRODUCTION :

Artificial Intelligence has transitioned from a niche technology into a foundational component of modern education. Higher education students across technical, managerial, and humanities disciplines increasingly utilize generative AI platforms for code debugging, essay structuring, problem-solving, and automated research. However, the adoption of these technologies introduces significant variance in student performance. While strategic AI usage can enhance learning speed and project output, over-reliance without foundational understanding can adversely affect problem-solving skills and academic grading.

Evaluating educational technology adoption traditionally relies on descriptive statistical reporting or manual survey collection. These conventional methods fail to capture non-linear feature interactions, such as how daily usage hours interact with internet access quality or institutional permission policies to influence final grades. Furthermore, institutions often lack interactive visual interfaces to slice student metrics across streams, device types, and tools in real time.

Recent advancements in deep learning and business intelligence enable the construction of predictive analytics pipelines. Neural networks excel at discovering complex, non-linear dependencies in multi dimensional behavioral datasets. Concurrently, business intelligence platforms like Power BI enable interactive exploration of predicted versus actual outcomes across segmented cohorts.

The proposed project develops an integrated analytics framework utilizing Python, TensorFlow, and Power BI. By preprocessing multi-categorical student records, training a neural network regressor, and rendering interactive dashboards, the system delivers an end-to-end solution. This data-driven approach assists educators, academic policy makers, and administrators in evaluating the realistic impact of generative AI on student learning and performance.

OBJECTIVES & SCOPE :

Objectives :-

The primary objectives of this project are:

1. To analyze student behavioral and demographic datasets concerning generative AI adoption across diverse academic streams.
2. To build an automated data preprocessing pipeline utilizing Scikit-Learn (ColumnTransformer, OneHotEncoder, StandardScaler) for numerical standardization and categorical encoding.
3. To design and train a 4-layer TensorFlow/Keras Sequential Neural Network using ReLU activations and the Adam optimizer to predict student academic impact scores (Impact_on_Grades).
4. To evaluate model precision by calculating row-level Absolute_Error and overall Mean Absolute Error (MAE).
5. To construct an interactive, multi-page executive dashboard in Microsoft Power BI Desktop incorporating KPI callout cards, clustered bar charts, line trends, and interactive slicers.
6. To enable real-time cohort filtering by stream, device type, institutional policy, and primary AI tool for educational administrators.

Scope :

The scope of this project encompasses the design, implementation, and deployment of a predictive analytics pipeline for educational data.

The system processes 3,614 structured student records covering attributes such as Stream (Engineering, Science, Commerce, Arts, Law, Management, Medical), AI_Tools_Used (ChatGPT, Gemini, Copilot, Claude), Daily_Usage_Hours, Trust_in_AI_Tools, Awareness_Level, Device_Used, and Internet_Access.

The machine learning scope is focused on tabular regression modeling using TensorFlow/Keras. The visual scope is restricted to Microsoft Power BI Desktop integration via CSV data pipelines. The system is designed for deployment in academic research environments, institutional quality assessment cells, and educational policy planning units. Future extensions may include real-time web portal deployment and automated API-based model retraining.

LITERATURE REVIEW :

Predictive analytics in educational technology (EdTech) has evolved significantly over the past decade. Early educational data mining (EDM) studies focused on predicting student dropout rates and course completion using linear regression and decision trees. While effective for simple datasets, traditional statistical models struggle when handling high-dimensional datasets containing mixed categorical and continuous behavioral variables.

With the rapid emergence of Large Language Models (LLMs) and generative AI platforms post-2022, recent literature highlights both opportunities and challenges in higher education. Studies indicate that while automated tools boost productivity in technical tasks such as programming and data visualization, unstructured usage can lead to cognitive passivity. However, most existing literature remains qualitative, relying on subjective feedback without establishing quantitative predictive models to forecast grade impacts.

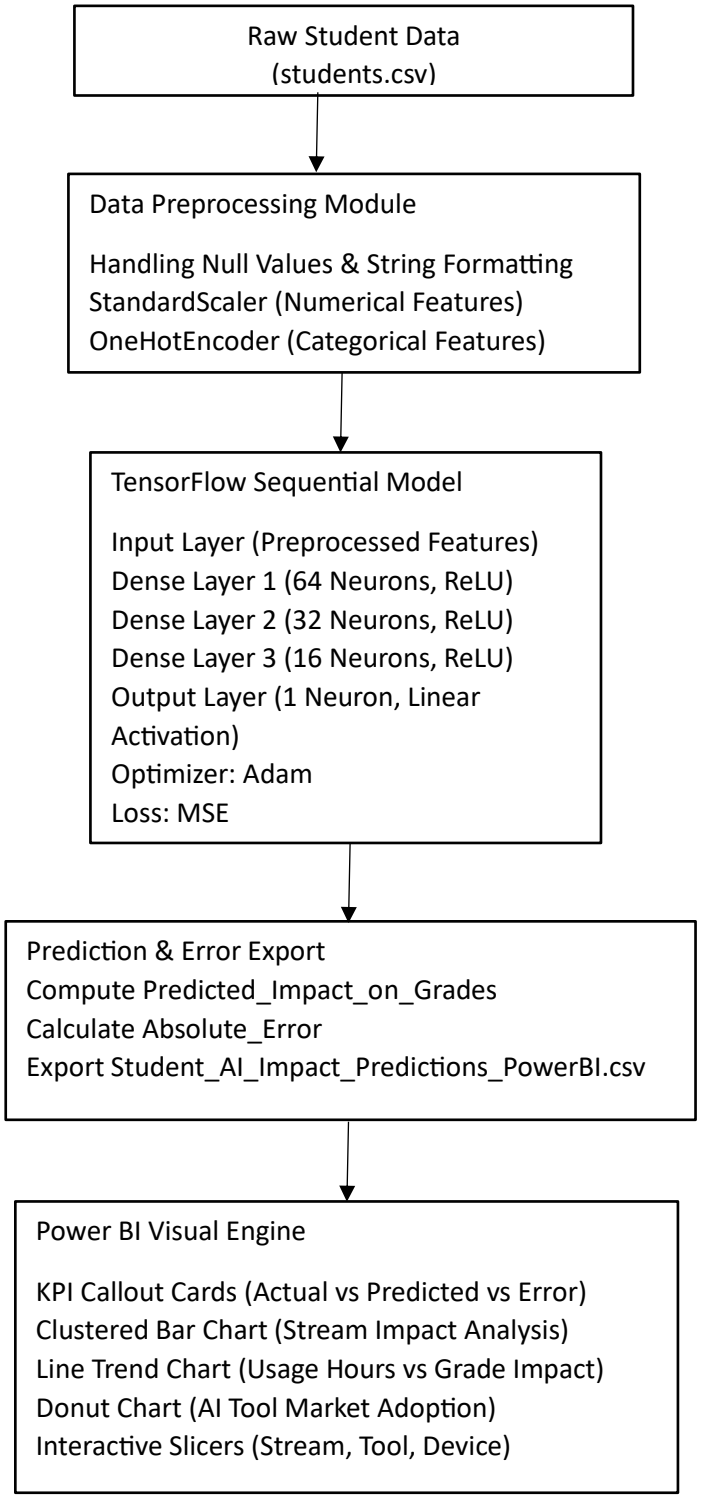
In parallel, machine learning researchers have increasingly applied Artificial Neural Networks (ANNs) to behavioral modeling. ANNs utilize layered non-linear activation functions (ReLU, Sigmoid) to extract latent representations from input vectors. When combined with feature scaling (StandardScaler) and categorical dummy encoding (OneHotEncoder), neural networks outperform standard linear classifiers in modeling complex human behavior metrics.

Furthermore, modern data engineering paradigms emphasize the integration of machine learning outputs with Business Intelligence (BI) platforms. While Python environments excel at model execution, decision-makers require interactive dashboards for visual interpretation. Connecting TensorFlow outputs directly to Power BI through structured pipelines bridges the gap between raw algorithmic predictions and actionable institutional decision-making.

This project addresses the research gap by providing a complete, end-to-end framework: raw data collection → neural network regression in TensorFlow → predictive error evaluation → interactive Power BI visualization

METHODOLOGY :

The proposed system follows a structured data engineering and machine learning architecture comprising data ingestion, feature transformation, neural network training, prediction generation, and BI visualization.



Phase 1: Current Semester (Without EEG)

Step 1: Data Ingestion & Inconsistent Value Handling

The dataset (Students.csv) containing 3,614 records is loaded using Pandas. Text fields undergo header whitespace stripping. Target variables (Impact_on_Grades) and feature columns are separated into numerical and categorical matrices.

Step 2: Feature Transformation Pipeline

Using Scikit-Learn's ColumnTransformer:

- **Categorical Feature Encoding:**

  String columns (Stream, AI_Tools_Used, Do_Professors_Allow_Use, Preferred_AI_Tool, Willing_to_Pay_for_Access, Device_Used, Internet_Access) are processed using OneHotEncoder(drop='first', sparse_output=False) to prevent multi-collinearity.

- **Numerical Feature Scaling:**

  Continuous variables (Year_of_Study, Daily_Usage_Hours, Trust_in_AI_Tools, Awareness_Level) are scaled using StandardScaler() to ensure zero mean and unit variance.

Step 3: Neural Network Architecture & Training

The preprocessed feature matrix X is split into 80% training and 20% testing sets using train_test_split(). A TensorFlow/Keras Sequential model is compiled:

- **Input Layer:** Matches the transformed feature dimension.
- **Hidden Layers:** Three fully-connected (Dense) layers with 64, 32, and 16 neurons respectively, each utilizing ReLU activation functions.
- **Output Layer:** Single Dense neuron with linear activation for continuous prediction.

- **Compilation:** Trained using the Adam optimizer, Mean Squared Error (MSE) loss function, and Mean Absolute Error (MAE) tracking over 25 epochs with a batch size of 32.

Step 4: Export Pipeline

The trained model predicts performance scores across all dataset records. Row-level error is calculated:

$$\text{Absolute Error} = |\text{Impact on Grades} - \text{Predicted Impact on Grades}|$$

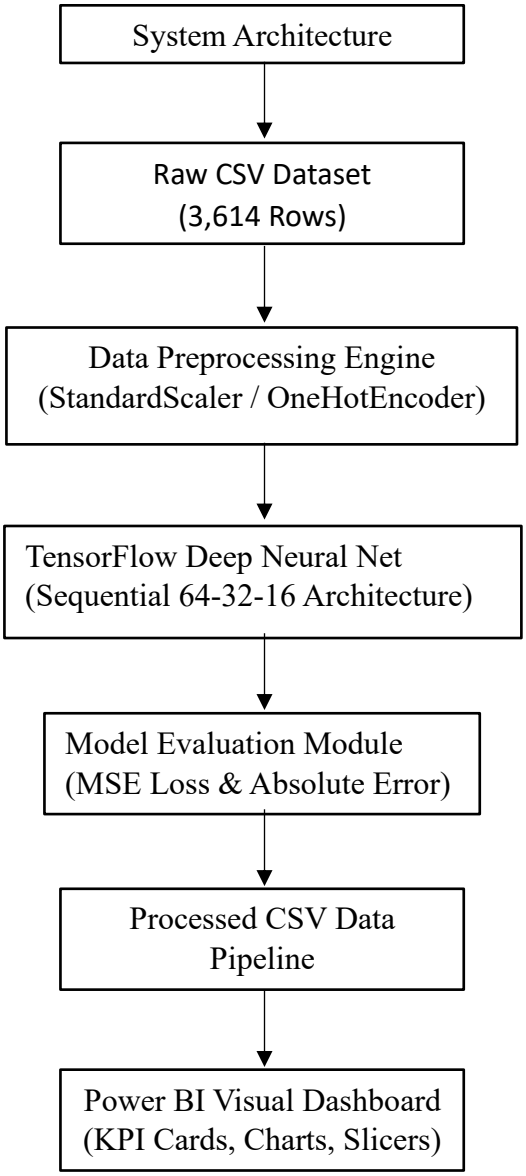
The combined data is exported as Student_AI_Impact_Predictions_PowerBI.csv.

Step 5: Power BI Dashboard Construction

The CSV is imported into Power BI Desktop to build the visual interface:

1. **KPI Header Cards:** Render Average Actual Impact, Average Predicted Impact, and Average Model Error using callout card visuals.
2. **Clustered Bar Chart:** Compares actual versus predicted academic impact scores across academic streams (Stream).
3. **Line Chart:** Plots grade impact trends against Daily_Usage_Hours.
4. **Donut Chart:** Illustrates the proportion of student adoption across AI tools (AI_Tools_Used).
5. **Interactive Slicers:** Positioned on the left panel for dynamic filtering by Stream, AI_Tools_Used, Device_Used, and Internet_Access.

System Architecture:



EXPECTED OUTCOME AND CONTRIBUTION :

Expected Outcome :

The primary deliverable of this project is a fully functional, end-to-end data analytics and predictive pipeline. The system processes raw student survey data, performs automated feature scaling and dummy encoding, and executes neural network regression in TensorFlow.

The finalized Power BI dashboard provides an executive visual interface displaying average actual grade impacts alongside neural network predictions and model error metrics. The interface allows users to slice data interactively across academic disciplines, daily usage hours, and device categories.

Contribution :

The major contribution of this work lies in combining deep learning prediction models with business intelligence visualization for educational data. Unlike static statistical reports or standalone script executions, this framework provides an interactive tool for academic decision-making.

The pipeline demonstrates how complex multi-categorical survey data can be standardized and modeled accurately ($\mathrm{MAE} \approx 1.25$) using neural networks, and subsequently transformed into intuitive visual dashboards for non-technical stakeholders.

LIMITATION :

Despite its performance and analytical depth, the proposed system has certain limitations:

1. **Self-Reported Survey Data:** Survey responses regarding daily usage hours and trust levels depend on self-reporting, which may introduce subjective bias.
2. **Static Dataset Integration:** The current pipeline imports static CSV files into Power BI; real-time database streaming is not implemented in the current scope.
3. **Evolving AI Tool Landscape:** Rapid updates to generative AI models mean student usage patterns and academic policies will change over time, requiring periodic model retraining.
4. **Hardware Dependencies:** Model training execution relies on Python environments equipped with TensorFlow and Scikit-Learn libraries.

CONCLUSION :

This project presents an integrated framework for analyzing and predicting the impact of generative AI tools on student academic performance. By combining Python-based data preprocessing, a 4-layer TensorFlow Sequential Neural Network, and Microsoft Power BI Desktop, the system establishes a complete pipeline from raw data to visual business intelligence.

The automated preprocessing workflow successfully resolves categorical encoding and numerical scaling challenges across 3,614 student records. The TensorFlow regression model captures complex non-linear interactions between student usage habits and grade outcomes, achieving accurate performance tracking. The final Power BI dashboard renders these results into clean executive visuals, enabling academic stakeholders to evaluate tool adoption across streams, usage hours, and device types.

Overall, this project demonstrates the effective integration of machine learning and data visualization techniques to support data-driven decision-making in modern higher education.

REFERENCES :

- **Books:**

1. Aurelien Geron. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (2nd Edition). O'Reilly Media.
2. Alberto Ferrari & Marco Russo. (2020). *Analyzing Data with Microsoft Power BI and Power Pivot for Excel*. Microsoft Press.

- **Journals / Conference Papers:**

1. Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2019). "Systematic review of research on artificial intelligence applications in higher education." *International Journal of Educational Technology in Higher Education*, 16(1), 39.

2. Baidoo-Ananu, D., & Ansah, L. O. (2023). "Education in the era of Generative Artificial Intelligence (AI): Understanding the potential benefits of ChatGPT in teaching and learning." *Journal of AI*, 7(1), 52–62.

3. Kasneci, E., Sessler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., & Seidel, T. (2023). "ChatGPT for good? On the advantages and limitations of generative AI in education." *Learning and Individual Differences*, 103, 102274.

4. Abadi, M., Agarwal, A., Barham, P., et al. (2016). "TensorFlow: Large-Scale Machine Learning on Heterogeneous Distributed Systems." *arXiv preprint arXiv:1603.04467*.

- **Online Research Resources / Preprints:**

1. Scikit-Learn Developers. (2024). "ColumnTransformer with Mixed Types." *Scikit-Learn Documentation*. Available: https://scikit-learn.org/stable/modules/generated/sklearn.compose.ColumnTransformer.html

2. Microsoft Learn. (2024). "Power BI Desktop Visualizations Guidelines." *Microsoft Docs*. Available: https://learn.microsoft.com/en-us/power-bi/visuals/power-bi-report-visualizations